

Assignment 8

Easy Booking

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1. Introduction

This report presents an analysis of the Apache web server logs generated by our Easy-Booking project website. The goal of the assignment was to evaluate how the website was accessed, identify usage patterns, and analyze errors recorded by the server.

We examined two types of logs:

1. **Apache access log** – contains all incoming HTTP requests.
2. **Apache error log** – contains warnings, failures, and invalid requests.

From these logs, we extracted statistics on:

- How often each page was accessed
- Who accessed the website (IP addresses)
- When access occurred (timestamps)
- Which browsers were used
- What errors occurred and when

With these two timeline diagrams were generated to visualize page access activity and error frequency over time.

2. Data Sources

The analysis is based on:

- access.log – containing client requests (IP, timestamp, page path, browser, status code)
- error.log – containing server-side warnings and error events

Additional clicktrails were manually created by team members to guarantee significant data volume because the publicly hosted project website had little natural visitation.

3. Method

The Apache logs were parsed and examined using a Python script.

The following actions are carried out by the script:

- Reads both log files line-by-line
- Extracts IP address, timestamp, requested page, HTTP status, and browser agent
- Builds structured datasets using pandas
- Aggregates statistics such as:
 - Page hit counts
 - Requests per IP
 - Browser usage
 - Error occurrences
- Exports summary tables as CSV files
- Produces two timeline charts using matplotlib:
 - Page Access Timeline
 - Error Timeline

All generated CSV files and diagrams are included in our repository under /hw8/ on github

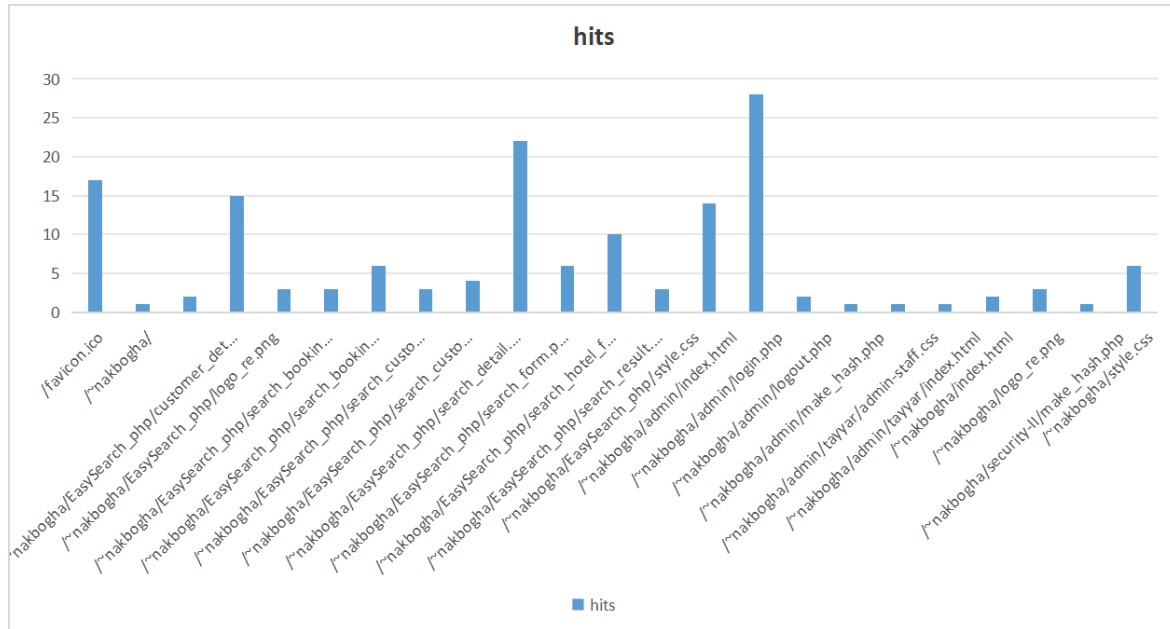
4. Results

4.1 Most Frequently Accessed Pages

Pages.csv is the source of the most popular pages.

These demonstrate how people interacted with different service components, mostly our team during testing.

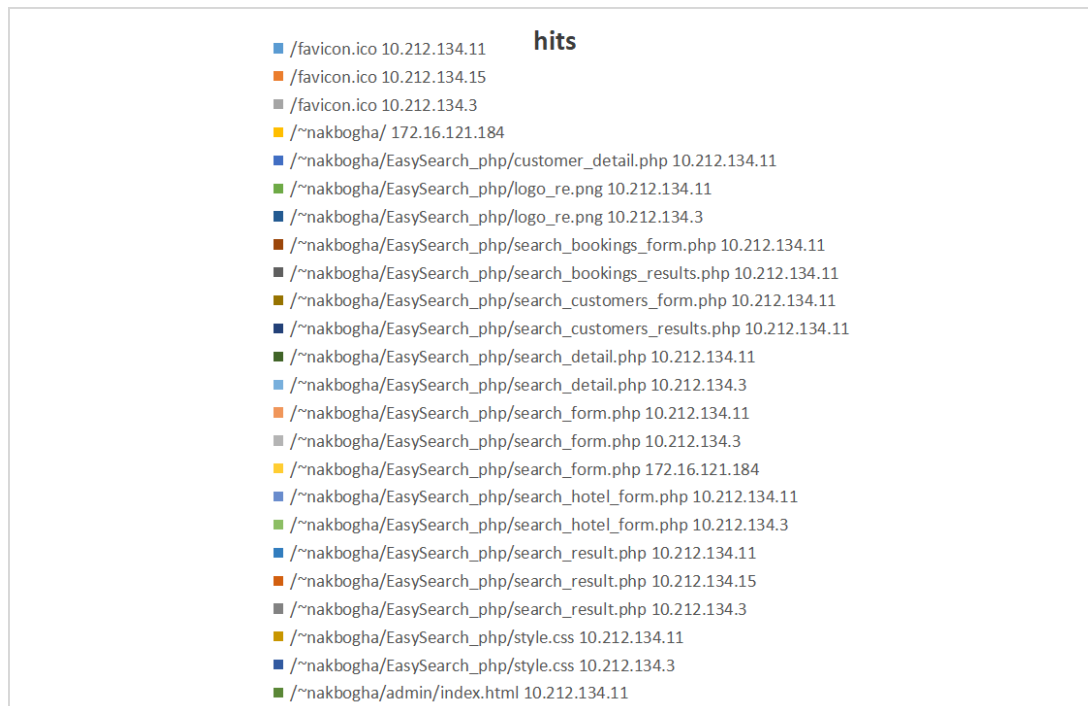
Due to frequent login and logout cycles throughout testing, the login page received the most requests. Pages pertaining to maintenance also displayed activity linked to insert/update operations.



4.2 Accesses by IP Address

Using pages_by_ip.csv, IP-based usage was analyzed.

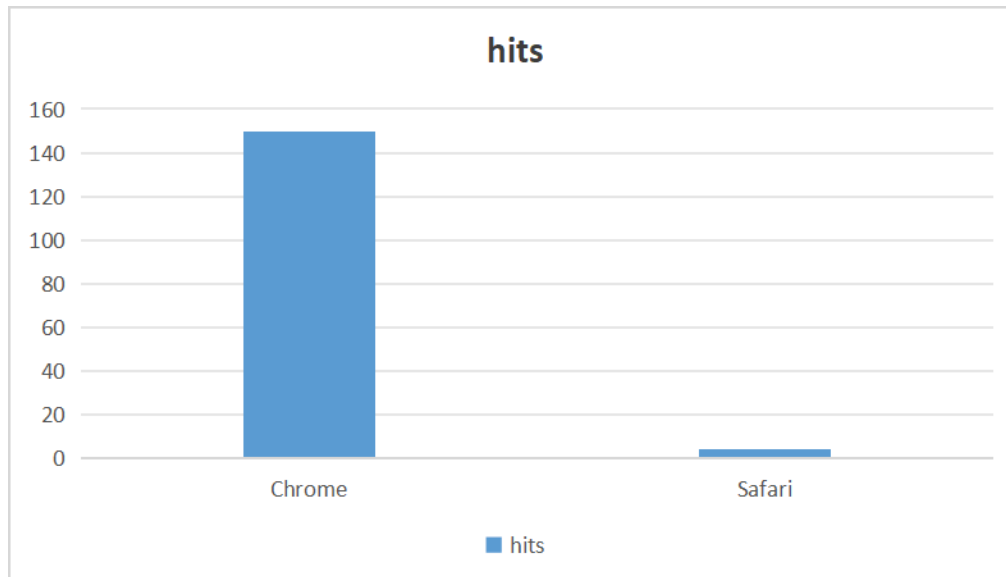
The majority of visits came from a small number of team members' internal IP addresses, which is consistent with manual test sessions.



4.3 Browser Usage

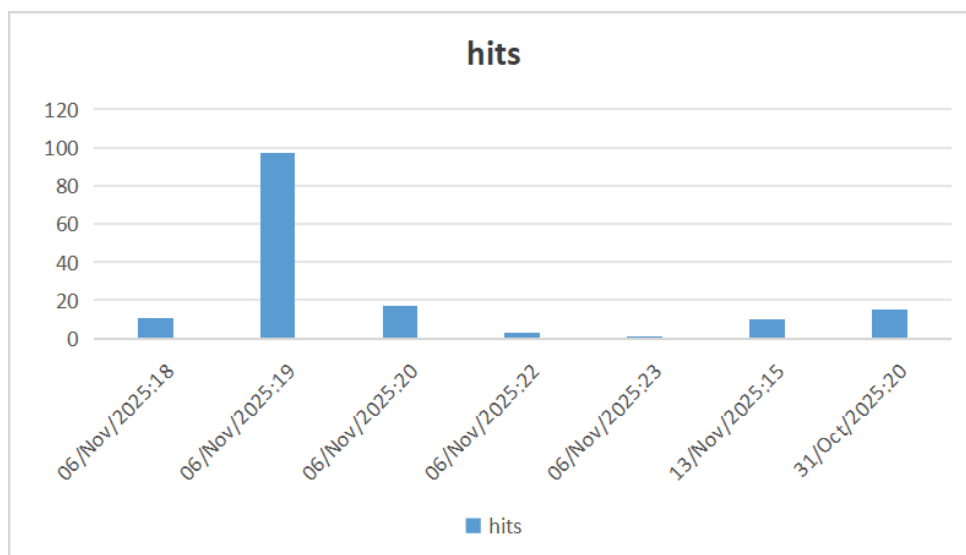
The browser statistics in browsers.csv show the distribution of user-agents.

Chrome was the most used browser, which is expected given typical usage among students.



4.4 Page Access Timeline

The timeline below (generated from timeline_hits.csv) shows the number of page hits per hour.

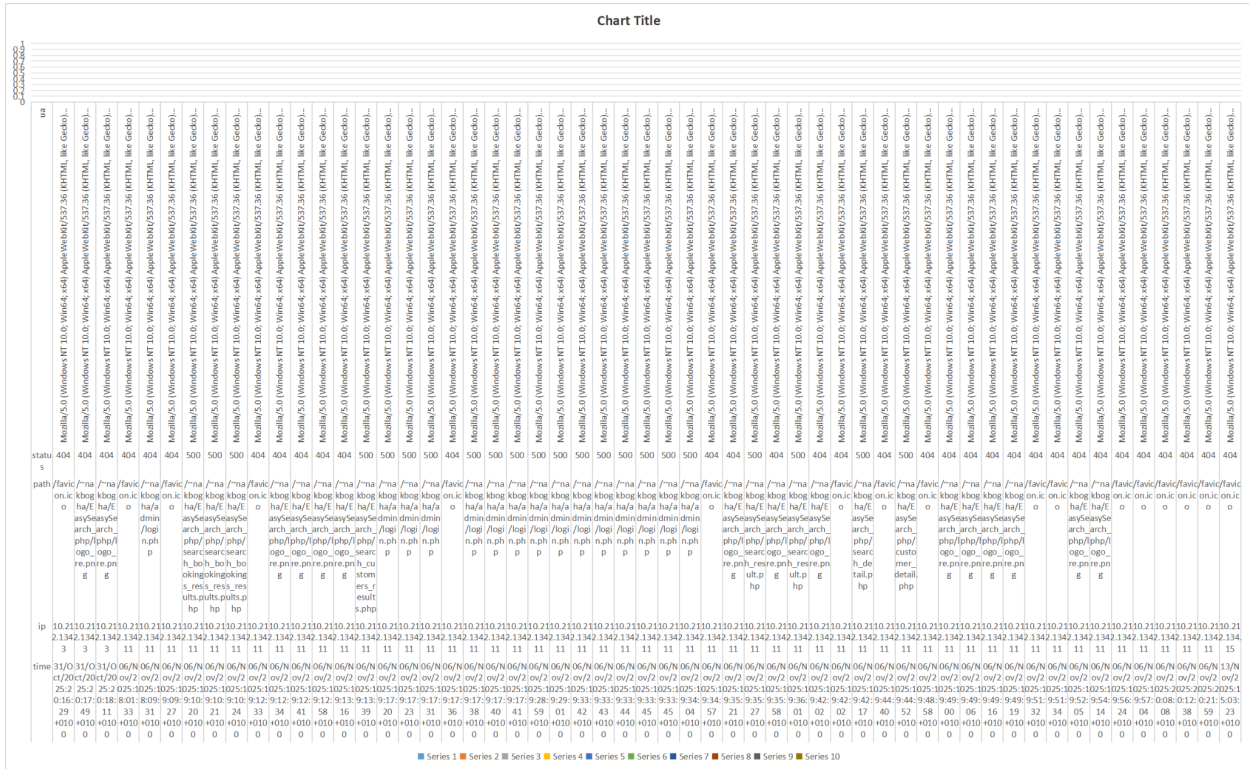


The activity spikes shown in this chart match to our manual testing schedule. Normal downtime is shown by times when there is little to no activity.

4.5 Error Summary

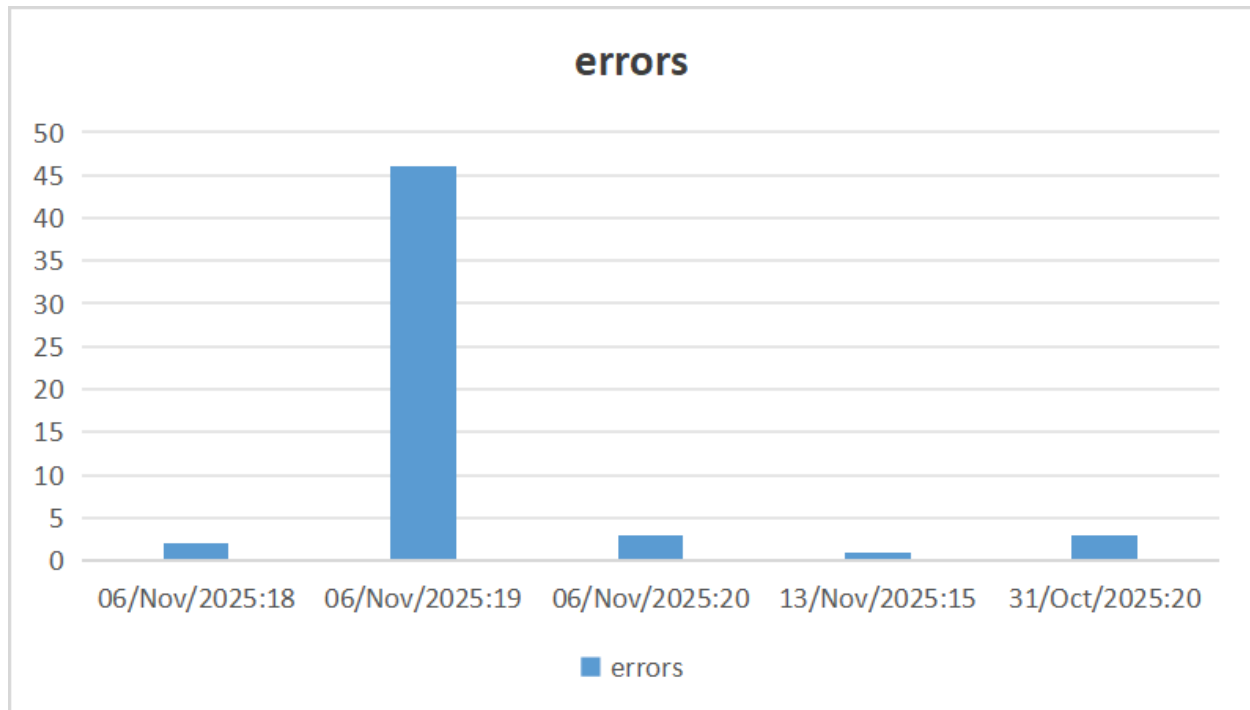
The URLs and timestamps linked to HTTP errors are displayed in error_entries.csv.

The majority of problems were 404 Not Found, which is usually caused by purposefully requesting sites that don't exist when debugging or developing.



4.6 Error Timeline

The second timeline shows the times of error events and is based on timeline_errors.csv.



Deliberate testing situations are reflected in the clustering of errors in particular time periods.

5. Tools Used:

- **Python 3.10**
 - **pandas** – structured data processing
 - **matplotlib** – timeline chart generation
 - **Regex** – Apache log parsing
 - **VS Code / PyCharm** – script editing
 - **Google Sheets / Excel** – quick verification of CSVs
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6. Conclusion

A detailed picture of how the Easy-Booking service was visited and evaluated is given by the web log review. The login page was the most accessed resource as a result of repeated authentication testing, according to the results.

- Every bit of traffic came from internal IPs that matched the devices of team members.
- The most popular browser was Chrome.
- During manual testing, the most common error was 404 Not Found.
- The activity peaks during testing sessions are clearly depicted in timeline diagrams.

The logs show both typical usage and controlled error-generation phases, and the web service generally performed reliably during testing.